

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Crankshaft Seal Replacer Kit, Cummins® Part Number 5394206, or equivalent
- Lubriplate™ 105, Part Number 3163086, or equivalent.

Additional Service Items

- Magnet
- Crocus cloth
- Clean, lint-free cloth.

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. See equipment manufacturer service information.
- Remove the fan and fan spacer(s). Refer to Procedure 008-040 in Section 8. (</qs3/pubsys2/xml/en/procedures/89/89-008-040.html>)
- Remove the fan belt. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/89/89-008-002.html>)
- Remove the fan belt tensioner on Generator-Drive applications. Refer to Procedure 008-087 in Section 8. (</qs3/pubsys2/xml/en/procedures/89/89-008-087-tr.html>)
- Remove the vibration damper and fan drive pulley. Refer to Procedure 001-052 in Section 1. (</qs3/pubsys2/xml/en/procedures/89/89-001-052.html>)

Remove

Use Crankshaft Seal Replacer Kit, Part Number 5394206, or equivalent, to remove the front crankshaft seal.

Apply Lubriplate™ 105, Part Number 3163086, or equivalent, onto the remover screw threads and tip (3).

Install the three replacer screws (1) into the crankshaft.

The crankshaft seal replacer (2) marked "OIL" on the top face is used for the removal of front dust seal and front oil seal as an assembly.

Install the remover screw (3) into the crankshaft seal replacer (2).

Slide the crankshaft seal replacer assembly (2) into the nose of the crankshaft. Push up to the front gear cover face.

Drill six self-tapping metal screws (4) into the front oil seal. Make sure the heads of the screws are flush on the face of the tool. Six screws **must** be used in this process to make sure the seal will come out evenly.

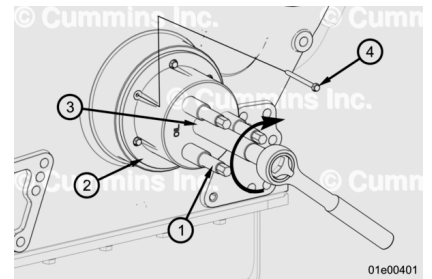
Use a wrench to rotate the remover screw (3) clockwise against the crankshaft front boss slowly, which slides the crankshaft seal replacer (2) along with seals outward.

The front seal and dust seal can be removed at the same time using the self-tapping metal screws included in the service kit.

Once the front seal and dust seal are removed from the crankshaft, remove and discard the self-tapping metal screws (4) to separate the front seal from the crankshaft seal replacer assembly (2).

Remove the three replacer screws (1) from the crankshaft.

Note : The front crankshaft seal is covered by a dust seal. Drill through this dust seal and remove it with the oil seal. Discard the old dust seal and oil seal.



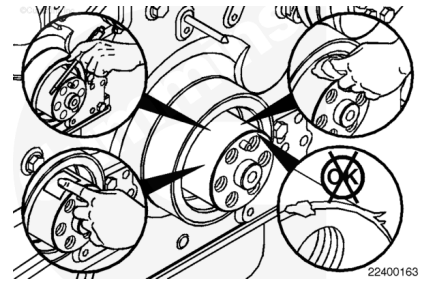
Clean and Inspect for Reuse

Use a magnet to remove any metal chips from the crankshaft and crankshaft seal bore in the front cover.

Use a crocus cloth to polish and remove deposits from the crankshaft. Visible scratches on the sealing surface can result in seal leakage.

Use a clean, lint-free cloth to clean the crankshaft and remove all traces of oil film.

Check the crankshaft end for nicks, burrs, and grooves.



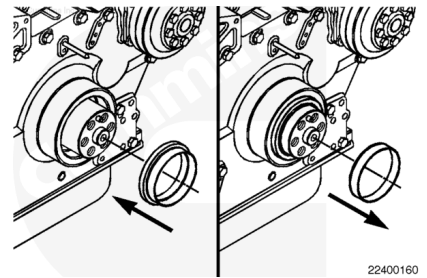
Install

⚠ CAUTION ⚠

Do not allow oil to come in contact with other areas of the seal to reduce the possibility of damage to sealing surfaces.

Using the plastic installation sleeve, if needed, place the new oil seal with the "OUT" marking facing outwards and sleeve assembly over the crankshaft nose and slide the seal toward the front cover. Make sure the seal is positioned squarely on the crankshaft.

Remove the plastic installation sleeve.



Use Crankshaft Seal Replacer Kit, Part Number 5394206, or equivalent, to install the front crankshaft oil seal.

Apply Lubriplate™ 105, Part Number 3163086, or equivalent, onto the installer screw threads and tip (3).

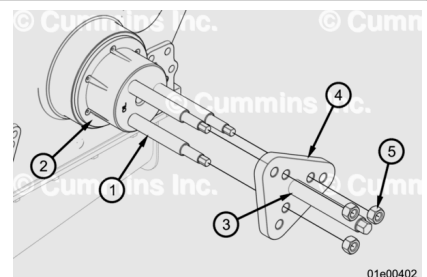
Install the three replacer screws (1) into the crankshaft.

The crankshaft seal replacer (2) marked "OIL" on the top face is used for the installation of the front oil seal.

Slide the crankshaft seal replacer (2) up to the nose of the crankshaft.

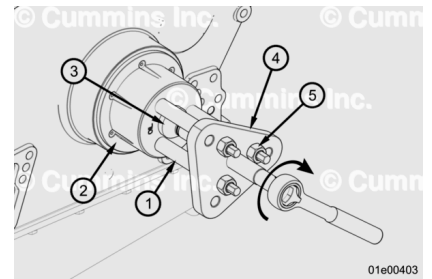
Install the replacer plate (4) with installer screw (3) onto the replacer screws (1).

Install the three regular hexagon jam nuts (5).



⚠ CAUTION ⚠

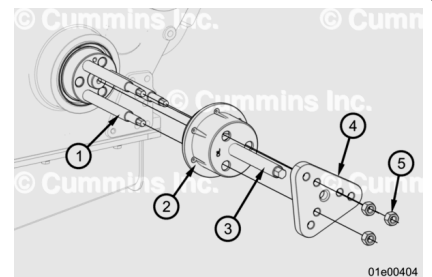
Do not use an impact wrench or air tools. Doing so can damage the tool.



Move the crankshaft seal replacer (2) toward the front seal by rotating the installer screw (3) **clockwise** until it is positioned against the front oil seal.

⚠ CAUTION ⚠

Do not overtighten the replacer screw assembly after the crankshaft seal replacer contacts the front cover. Doing so can damage the tool.

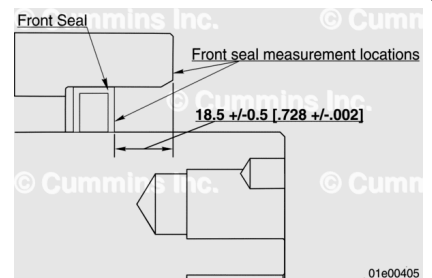


Holding the crankshaft seal replacer (2), rotate the replacer screw (3) **clockwise** until the crankshaft seal replacer is positioned against the front cover.

Remove the crankshaft seal replacer from the replacer screw assembly.

Measure the front oil seal depth from the front gear cover face.

Front Oil Seal Depth from Front Gear Face		
mm		in
18	MIN	0.709
19	MAX	0.748



If the front oil seal depth is **not** within specifications, the front oil seal **must** be replaced.

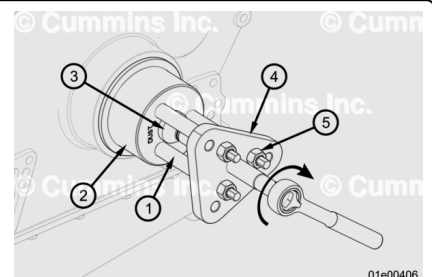
Note : The dust seal must be ordered separate from the front seal.

Place the dust seal over the crankshaft nose and slide toward the front cover. Make sure the dust seal is positioned squarely on the crankshaft. The dust seal groove **must** face the operator.

Use Crankshaft Seal Replacer Kit, Part Number 5394206, or equivalent, to install the front crankshaft dust seal.

Install the three replacer screws (1) into the crankshaft.

The crankshaft seal installer (2) marked "DUST" on the top face is used for the install of front dust seal.



Slide the crankshaft seal installer (2) up to the nose of the crankshaft.

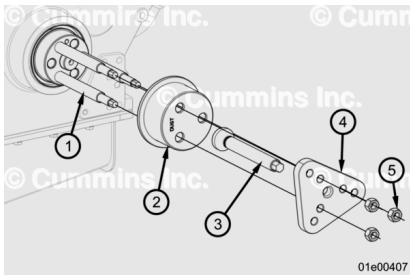
Install the replacer plate (4) with installer screw (3) onto the replacer screws (1).

Install the three regular hexagon jam nuts (5).

Move the crankshaft seal installer (2) toward the dust seal by rotating the installer screw (3) **clockwise** until it is positioned against the front dust seal.

Holding the crankshaft seal installer (2), rotate the installer screw (3) **clockwise** until the crankshaft seal replacer is positioned against the front cover.

Remove the crankshaft seal installer from the replacer screw (1) assembly.



Measure the front dust seal protrusion from the front gear cover face.

Front Dust Seal Protrusion from Front Gear Face		
mm		in
2.25	MIN	0.089
1.75	MAX	0.069

If the front oil seal depth is **not** within specifications, the front oil seal **must** be replaced.

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the vibration damper and fan drive pulley. Refer to Procedure 001-052 in Section 1. (</qs3/pubsys2/xml/en/procedures/89/89-001-052.html>)

- Install the fan belt tensioner. Refer to Procedure 008-087 in Section 8.
(/qs3/pubsys2/xml/en/procedures/89/89-008-087-tr.html)
- Install the fan belt. Refer to Procedure 008-002 in Section 8.
(/qs3/pubsys2/xml/en/procedures/89/89-008-002.html)
- Install the fan spacer(s) and fan. Refer to Procedure 008-040 in Section 8.
(/qs3/pubsys2/xml/en/procedures/89/89-008-040.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for leaks.

Last Modified: 14-Jul-2021
